

# Utkarsh Shaning

LinkedIn | GitHub | Portfolio | LeetCode

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## EDUCATION

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- **Jaypee University of Information Technology** Solan, Himachal Pradesh  
*Bachelor of Technology in Computer Science; CGPA: 7.79* 2024 – 2028
  - Relevant Coursework: Database Management Systems (DBMS), Object-Oriented Programming (OOPS), Data Structures and Algorithms

## PROJECTS

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- **Distributed Task Queue System** *GitHub | Live*
  - Tech Stack: Java, Spring Boot, RabbitMQ, Docker, AWS EC2
  - Built a high-throughput, distributed message queue for handling asynchronous background tasks efficiently across multiple microservices.
  - Implemented fair-dispatch load balancing across multiple worker nodes using RabbitMQ prefetch and manual ACK, ensuring zero message loss under heavy traffic.
  - Engineered a Dead Letter Queue with exponential backoff retry mechanism for fault tolerance.
  - Deployed on AWS EC2 and automated recurring task execution using cron jobs, reducing manual intervention for scheduled background jobs.
- **Scalable Microservices-Based Salon Booking Platform** *GitHub*
  - Tech Stack: Spring Boot, RabbitMQ, React.js, MongoDB, API Gateway, Stripe, Keycloak, Docker
  - Designed RESTful microservices using layered architecture (Controller–Service–Repository) ensuring modularity and separation of concerns.
  - Implemented service discovery with Eureka Server and centralized routing and filtering via API Gateway.
  - Developed asynchronous event-driven workflows using RabbitMQ to decouple booking, notification, and payment services.
  - Integrated Stripe webhooks for real-time transaction handling and synchronized MongoDB updates across services.
  - Secured distributed services using Keycloak with JWT-based authentication and RBAC.
  - Enabled resilient inter-service communication using OpenFeign clients with circuit breaker mechanisms.
  - Containerized all services using Docker Compose for local orchestration and deployment.
- **askPDF** *GitHub | Live*
  - Tech Stack: FastAPI, LangChain, ChromaDB, Python, Streamlit
  - Developed RAG-based architecture for contextual question answering over unstructured PDF documents.
  - Implemented document chunking, embedding generation, and semantic similarity search using LangChain and ChromaDB.
  - Exposed inference through FastAPI REST endpoints and built UI for document ingestion and conversational querying.

## TECHNICAL SKILLS

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- **Languages:** Java, Python  
**Frameworks:** Spring Boot, Spring MVC, FastAPI, LangChain  
**Databases:** MongoDB, MySQL  
**Messaging & Distributed Systems:** RabbitMQ  
**Security:** Keycloak, JWT, RBAC, Spring Security  
**Dev Tools:** Docker, Postman, Swagger/OpenAPI, Git, Streamlit

## HONORS & AWARDS

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- Won **3rd Position (Innovation Award)** at **Cyber AI Hackathon 2025** (International – University of Derby, UK, supported by American Society of Engineers); competed among 100+ teams, drove backend development and AI logic for the winning solution. *Live | GitHub*